

CNN Training on CIFAR-10 Dataset

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1 Introduction

This experiment investigates the performance of a Convolutional Neural Network (CNN) trained on the CIFAR-10 dataset. The objectives of this lab include designing a CNN model, computing FLOPs, visualizing gradient and weight updates, and reporting all results using Weights & Biases (WandB).

2 Dataset

The CIFAR-10 dataset consists of 60,000 color images of size 32×32 belonging to 10 different classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The dataset is split into 50,000 training images and 10,000 test images.

3 Model Architecture

The implemented CNN model consists of:

- Two convolutional layers with ReLU activation and max pooling
- Fully connected layers for classification

The model takes an input of size $3 \times 32 \times 32$ and outputs probabilities for 10 classes.

4 FLOPs and Parameters

The FLOPs and number of parameters were computed using the `thop` library in PyTorch.

- FLOPs: Reported in WandB logs
- Parameters: Reported in WandB logs

5 Training Setup

- Optimizer: Adam
- Loss Function: Cross Entropy Loss
- Batch Size: 128

- Epochs: 30
- Learning Rate: 0.001

6 Results

6.1 Training Loss



Figure 1: Training loss vs epochs (from WandB)

6.2 Training Accuracy

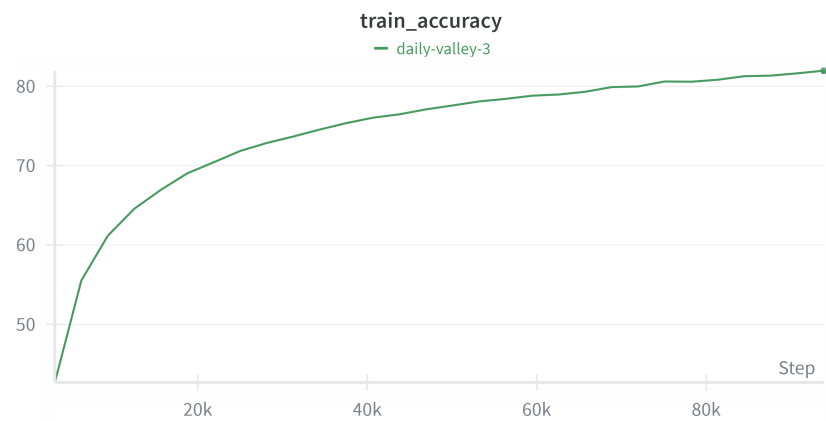


Figure 2: Training accuracy vs epochs (from WandB)

7 Observations

- The training loss decreases steadily over epochs, indicating effective learning.

- Training accuracy improves with epochs and stabilizes after around 20 epochs.
- Gradient visualizations indicate stable gradient flow without vanishing or exploding gradients.
- Data augmentation improves generalization performance.

8 WandB and GitHub Links

All visualizations and experiment logs are available at:

WandB: <https://wandb.ai/ravisharma112-prom-iit-rajasthan/cifar10-cnn-lab/workspace?nw=nwuserravisharma112>

GitHub Repository: https://github.com/Ravi110296/MLOps-Ravi_Sharma-M25CSA024/tree/Ravi_Sharma_M25CSA024_lab2_worksheet

9 Conclusion

In this lab experiment, a CNN was successfully trained on the CIFAR-10 dataset. FLOPs were calculated, and gradient and weight update flows were visualized using WandB. The results demonstrate stable convergence and effective learning behavior of the network.